

PUNIT MISHRA

Senior ML & AI Tech Lead | Agents, Inference & Multi-Tenant Platforms at Enterprise Scale

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PROFILE

I like taking machine learning the whole distance — from rough prototype to something a real customer can lean on — and I've done it at enterprise scale for 14+ years. As Senior ML & AI Tech Lead in SAP CX Data & AI I *lead a team of thirteen* while staying hands-on: I **architect the systems and also write, deploy, and own them end to end**, from agent runtimes and LLM inference pipelines to **GPU serving on custom bare-metal H100 clusters**. Recently I brought **SAP's Nova model into our natural-language-to-SQL stack** for cost control and model optionality, and architected a **hardened internal-auth contract** that closes a cross-tenant data-leak class. I care about the "so what": automated onboarding over manual tenant setup, latency cut on our flagship retail deployment, security you can actually verify. Deep in **Python** and **Rust**, with multi-model serving, distributed inference, vector search, and **secure, ethical AI**.

EXPERIENCE

SAP Sep 2018 – Present
Senior ML & AI Tech Lead | CX Data & AI San Ramon, CA

- I brought **Nova**, SAP's own code LLM, into our **natural-language-to-SQL** stack (S/4HANA + Ariba) as a first-class alternative to Claude — self-hosted serving, a retriever-reranker, and seven benchmark lanes — **cutting dependence on external frontier models** and lifting decision-match **59.2% → 64.8%**; NL-to-SQL ships in SAP's **2026 public and private cloud releases**.
- I architected the **hardened internal-auth contract** (ADR 0021) — a constant-time shared-secret check before any tenant header is trusted, canonical tenant resolution, and **IAS alongside XSUAA** — shipped backward-compatibly across four services to **close a real cross-tenant data-leak class**.
- I turned APEX from manual, XSUAA-only tenancy into **direct app tenancy auto-provisioned via BTP UCL formations, enabling self-service customer onboarding** — and serve as **forward-deployed engineer** bringing up our first production enterprise customer, a **Fortune 500 retailer**.
- I rewrote the legacy Shopping Assistant's **vision** (Vision-DNA, object detection, visual search) into first-class agent services behind a unified **Agent Gateway** and an **MCP Hub**, turning dead-end legacy code into reusable, discoverable services.
- I **lead a team of thirteen** and set technical direction through the ADRs I author, our SDLC and engineering standards, and **AI ethics & compliance** as an **SAP Security Expert**.

CALLIDUSCLOUD Feb 2014 – Sep 2018
Senior Software Engineer | Thunderbridge AI / SAP Sales Cloud SF Bay Area (acquired by SAP, \$2.4B)

- Designed and built the **Thunderbridge AI** backend in Python on **Dataiku**, with multi-tenancy running every customer workload for SAP Sales Cloud optimization.
- Ported the full Dataiku ML stack to **GKE on-prem (Google Anthos)** with Istio, and built the CI/CD backbone — Jenkins on Kubernetes, Kaniko image builds, and Artifactory distribution.
- Built developer-productivity tooling** — automated pull-request merges with Palantir Bulldozer and Policy-Bot on Kubernetes, wired into SAP GitHub Enterprise via OAuth to capture all Git events — cutting manual merge overhead across the team.

IBM Oct 2012 – Dec 2013
Automation Systems Engineer Dallas–Fort Worth, TX

- Built a reporting and **data-validation automation framework** for a major US bank.
- Developed reusable **Selenium / Java** test APIs (JUnit, TestNG) adopted across the firm.

EARLIER

UC Berkeley Law IST — Web Developer (2011) | **LawPivot (Google Startup Labs)** — SWE Intern (2011) | **LawPivot** — SWE Intern (2009–2010)

SKILLS

AI & AGENTS	Agent orchestration (LangGraph, LangChain), Agent Gateway , MCP Hub (registry / discovery), LLM serving, fine-tuning & evaluation, NL-to-SQL, RAG, retriever-reranker, PyTorch, OpenAI / CLIP, automated eval harnesses & benchmark lanes
GPU & SERVING	Ray (distributed inference / eval engines), custom bare-metal H100 clusters, multi-model serving (self-hosted Nova + Claude / Qwen / Granite-class), self-hosted LLM / embedding serving, Knative / KServe
INFRA & CLOUD	Kubernetes, Kyma / Gardener, Istio, Argo / GitOps, Jenkins, Docker, CI/CD; vector DBs (Qdrant, FAISS), Redis; Azure (AKS), GCP (GKE / Anthos)
PLATFORM & DATA	gRPC / protobuf, IAS / XSUAA (OIDC / PKCE), CAP / cds-mtxs, BTP UCL formations; PostgreSQL, SAP HANA, Oracle, Vertica, HDFS / Spark, Kafka
LANGUAGES	Python (FastAPI, Django), Rust , Java, Scala, TypeScript / Next.js
DEV PRODUCTIVITY	GitHub Enterprise automation (Bulldozer, Policy-Bot), Artifactory, Kaniko, ADRs, ephemeral PR environments
SECURITY & ETHICS	SAP Security Expert; internal-auth / identity design, threat modeling, SAST / SCA (Coverity, Black Duck, Fortify); AI ethics, safety & compliance

EDUCATION

UNIVERSITY OF CALIFORNIA, BERKELEY — B.S. Electrical Engineering & Computer Science

2012